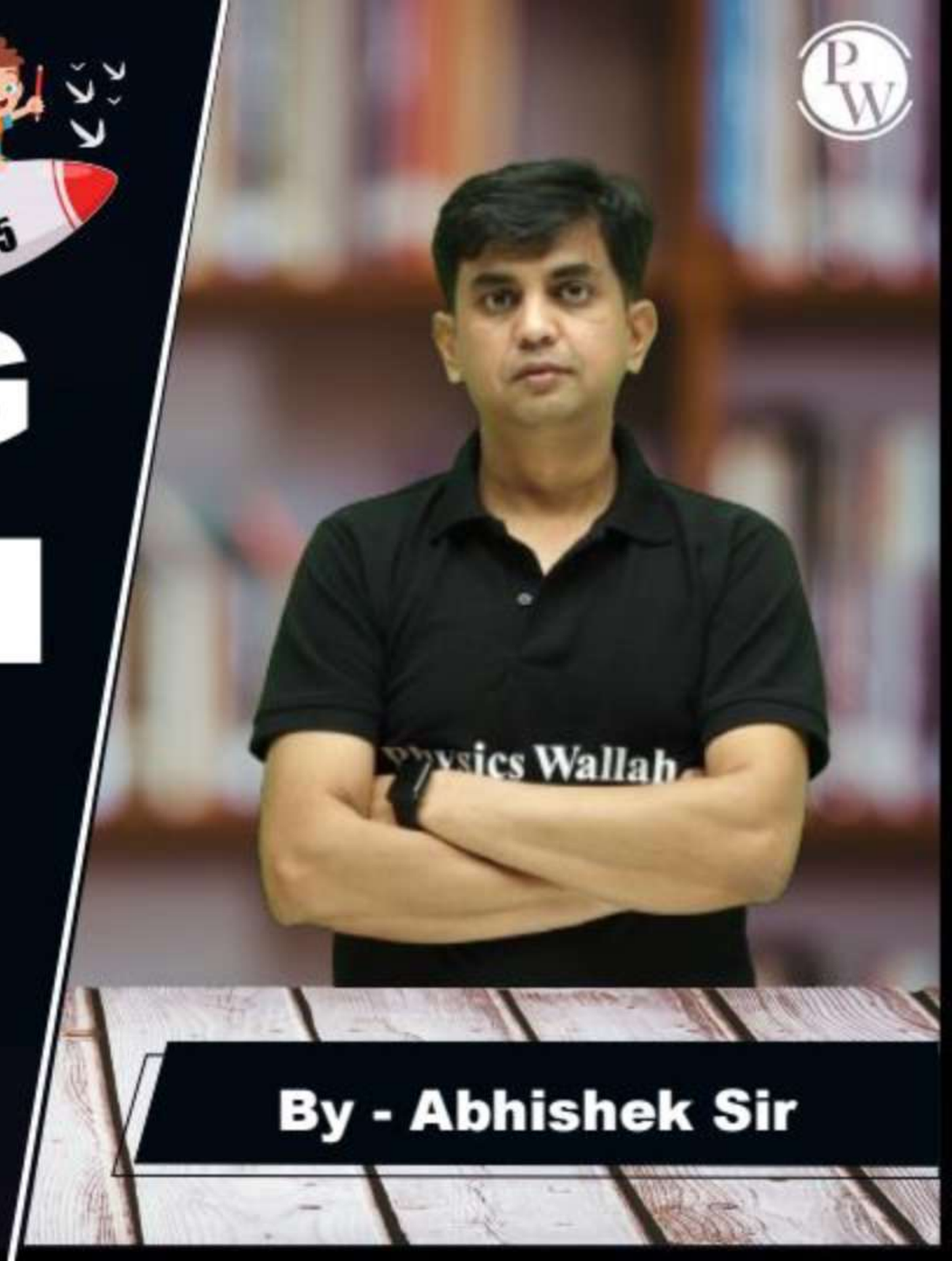
CS & IT ENGINEERING

Computer Network

Application Layer

Lecture No. - 02



By - Abhishek Sir

Recap of Previous Lecture

Topic Application Layer Protocol

Topic DNS

Topics to be Covered

Topic DNS

Topic HTTP

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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DNS Hierarchy:-

1) Root Name Server
2) TLD Name Server
3) Authoritative Name Server

Topic : Local DNS Name Servers

→ When host make DNS query, it is sent to its local DNS server

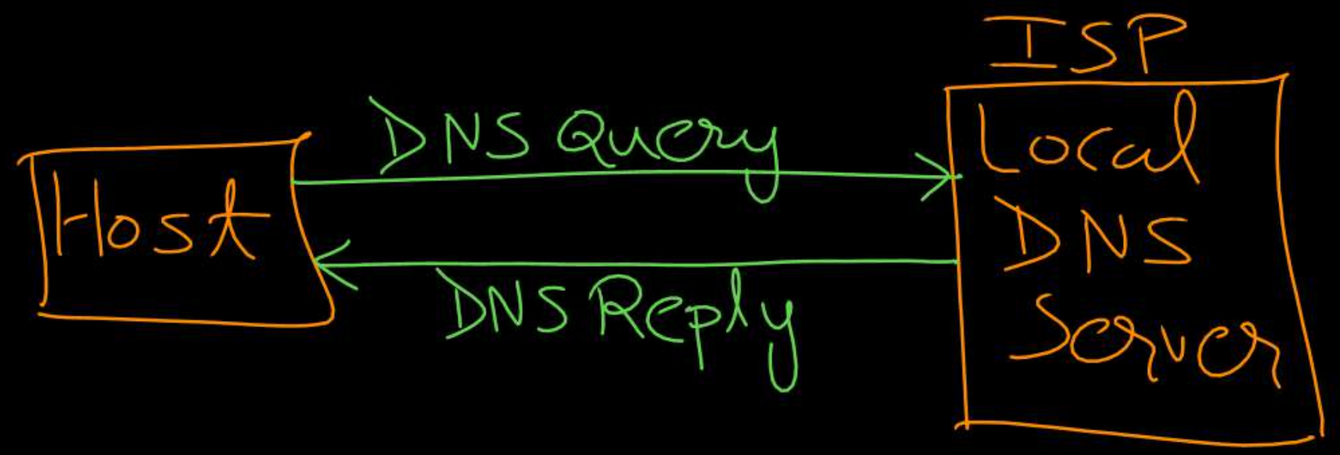
→ Local DNS server returns reply

→ if Local DNS server does not have valid name-address pair then it forward request into DNS hierarchy for resolution

Topic : Local DNS Name Servers

→ Each ISP has local DNS name server

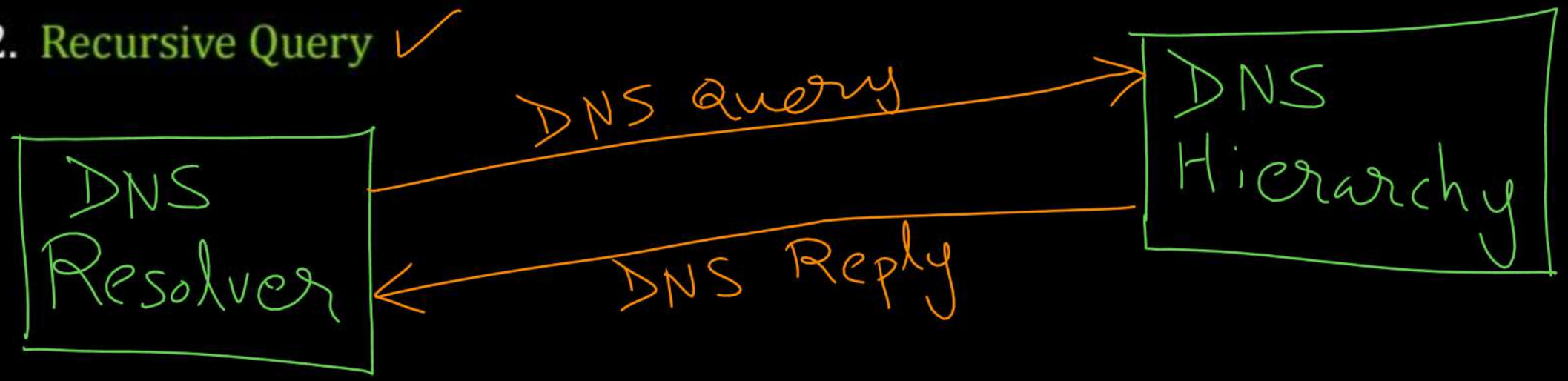
→ Local DNS server does not belongs to DNS hierarchy



Topic : DNS Name Resolution

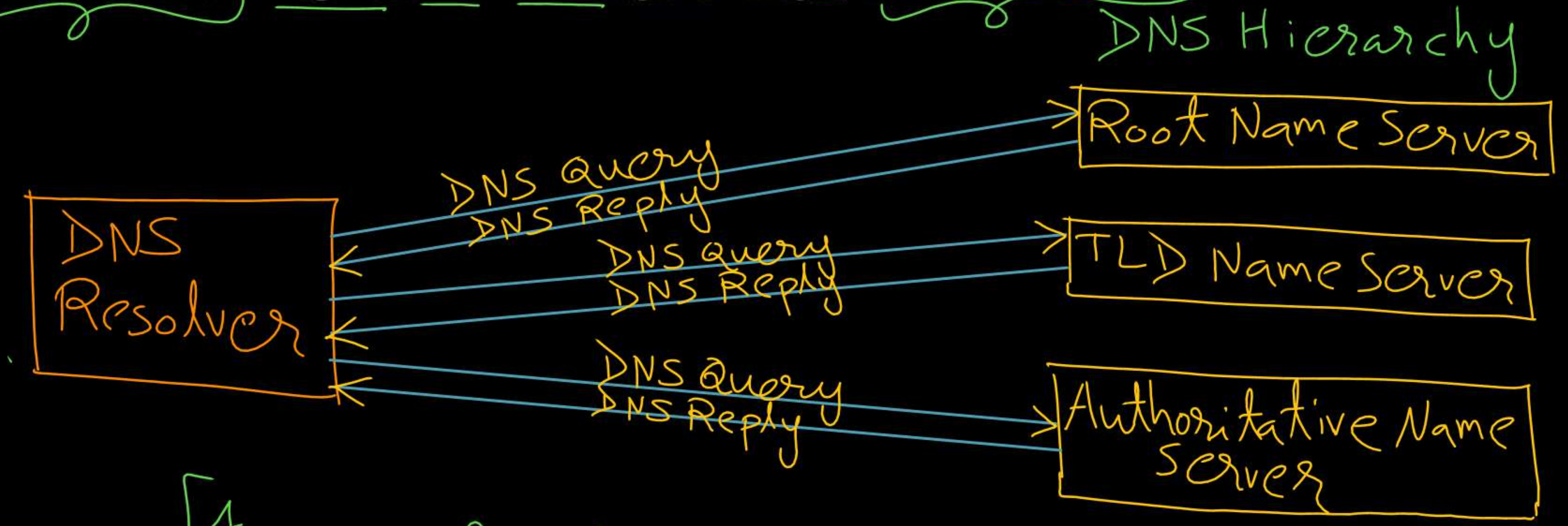
→ DNS Name Resolution or Lookup : (process)

1. Iterated Query ✓
2. Recursive Query ✓



Topic : Iterated Query

→ Contacted server replies with name (address) of server to contact



[In case of 3-tier Hierarchy
Total 3 RTT required for Name Resolution]

Domain Name : www.google.com

It is 3-tier

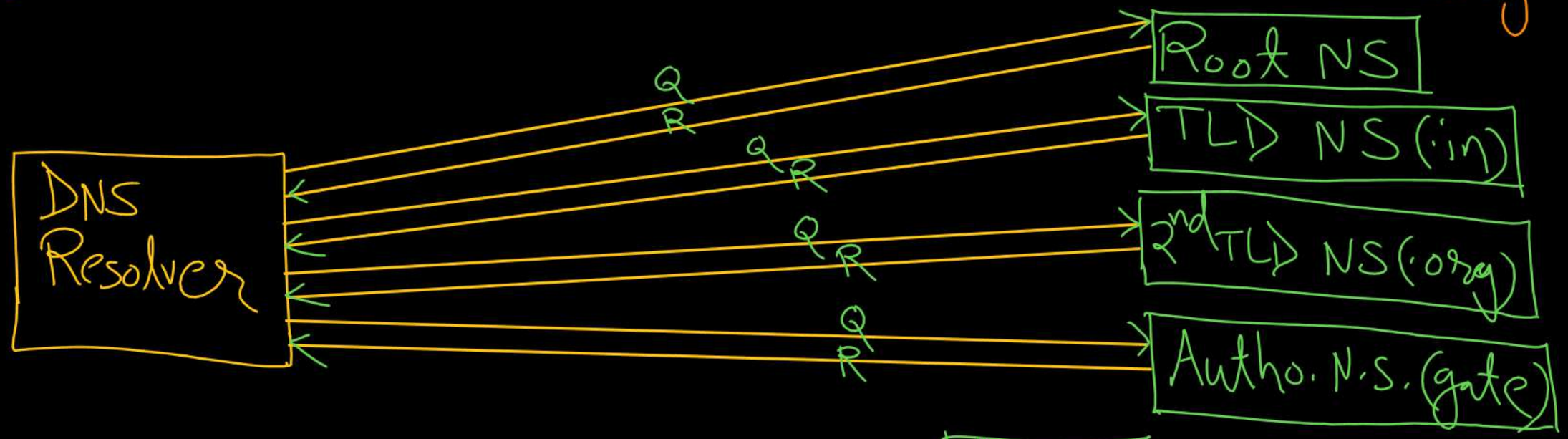
#Q. Consider the resolution of the domain name www.gate.org.in by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.

IITKGP

[GATE-2022, 1-Mark]

DNS Hierarchy

Query ⇒ DomainName : www.gate.org.in

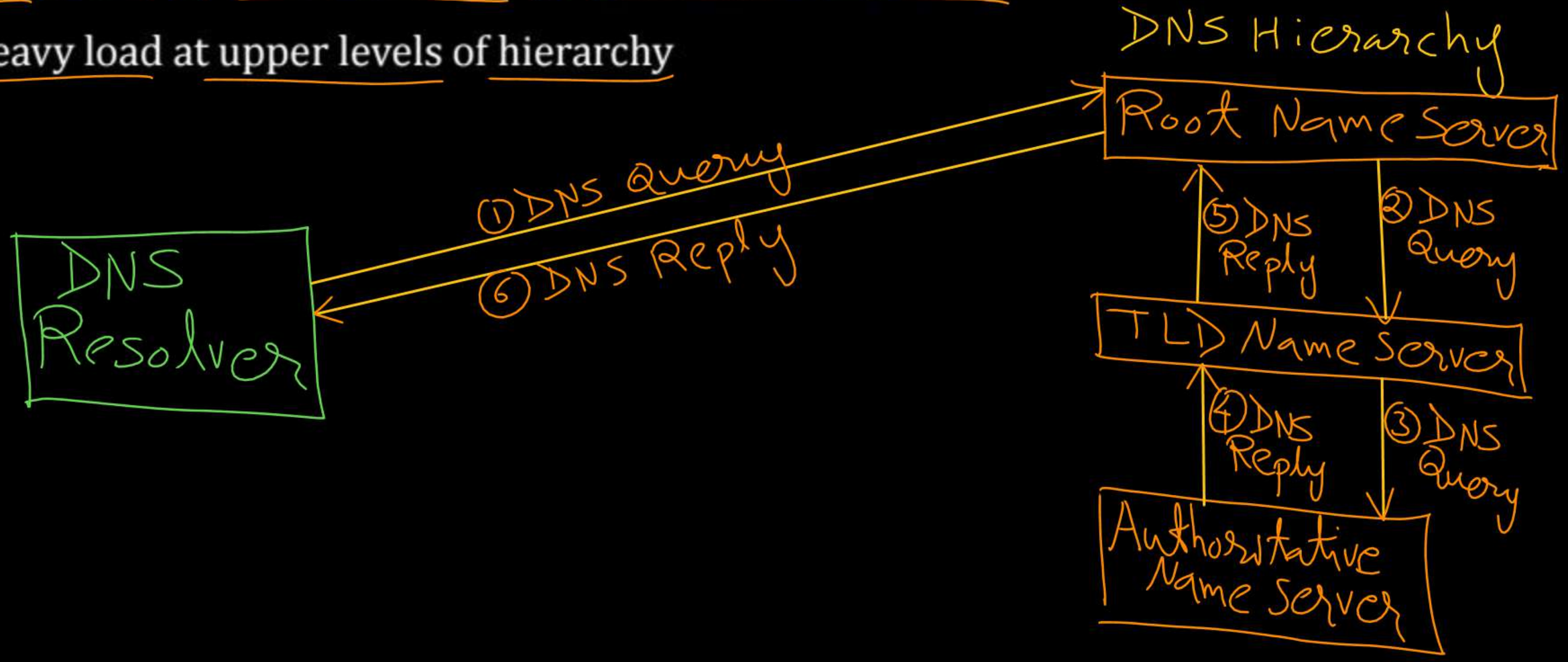


Ans: 4

Topic : Recursive Query

→ Put burden of name resolution on Contacted name server

→ Heavy load at upper levels of hierarchy



Topic : DNS Caching

→ Once (any) name server learns mapping, it caches mapping

→ Immediately returns a cached mapping in response to a query

=> Caching improves response time

=> Cached entries time-out (disappear) after some time (TTL)

Topic : DNS Records

→ Distributed database storing resource records (RR)

→ RR format : (Name, Value, Type, TTL)

Type	Name	Value
A	Host Name	IP Address
NS	Domain Name	Authoritative Name Server IP Address
CNAME	Alias Name	Canonical Name
MX	Name	Associated SMTP Mail Server Name

Topic : DNS Message Format

→ DNS query and reply message have same format

Identification	Flags	12 bytes
Number of questions	Number of answer RRs	
Number of authority RRs	Number of additional RRs	
Questions (variable number of questions)		Name, type fields for a query
Answers (variable number of resource records)		RRs in response to query
Authority (variable number of resource records)		Records for authoritative servers
Additional information (variable number of resource records)		Additional "helpful" info that may be used

Topic : DNS

→ DNS uses UDP as transport protocol for query and reply message when message size is less than equal to 512 bytes

→ if DNS messages size is greater than 512 bytes then DNS uses TCP as transport protocol

→ if DNS client doesn't get response from DNS server for its query then DNS uses TCP as transport protocol for query and reply message

→ DNS uses TCP for 'zone transfer'

P_1

Port = X

TCP Socket

UDP Socket

(IP
+ port
+ protocol)

Topic : HTTP

→ Web page consists of objects

→ Each objects can be stored on different web servers

→ Object can be HTML file, JPEG image, Java Applet, Audio File etc

→ Web page consists of base HTML file which includes several referenced objects [Each addressable by a URL]

Topic : HTTP

→ Hypertext Transfer Protocol : HTTP

→ Web's application layer protocol

→ Client - Server Model

HTML: HyperText Markup Lang.

Topic : HTTP

Client :

→ Browser that requests, receives, (using HTTP protocol) and display web objects

Server :

→ Web server sends (using HTTP protocol) objects in response to requests

Topic : HTTP uses TCP

→ Client initiates TCP connection (creates socket) to server on Port 80

→ Server accepts TCP connection from client

→ HTTP messages (application layer protocol messages) exchanged between browser (HTTP client) and web server (HTTP server)

→ TCP connection closed

Topic : HTTP is stateless

→ Server maintains no information about past client requests

→ Protocols that maintain "state" are complex
[Past history (state) must be maintained]

Topic : HTTP Connections

→ Two types of HTTP connections :

1. Non-persistent HTTP : HTTP/1.0
   → At most one object sent over TCP connection
   → Downloading multiple objects required multiple connections

2. Persistent HTTP : HTTP/1.1
   → Multiple objects can be sent over single TCP connection [between Client and Server]

Topic : Round Trip Time

RTT : Round Trip Time

→ Time for a small packet to travel from client to server and back

Round Trip Propagation Time < Round Trip Time

2 mins Summary

Topic DNS

Topic HTTP

THANK - YOU